

Performance of different Machine learning methods on predicting the result of a football match

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1 Abstract

Predictions for football matches have always been a notoriously difficult task because of the three way outcome of a match, also called 1X2 betting. What is especially difficult is to determine the cases where neither team has the edge, therefore predicting a draw. In this paper, different machine learning methods are applied to tackle this task and to see whether some are better suited for this problem than others. The dataset used for predictions consists of historical records, specifically the last 12 seasons of the English Premier League (2014-2026). The dataset was enriched with additional features that were derived from the existing ones, such as rolling averages of the features, points per game, team elo ranking and expected goals (xG). The models were evaluated on their log loss and brier score. The models used could be divided into discriminative and generative. The discriminative models included a feed-forward neural network, ensemble methods as Random forest and Gradient boosting, multinomial Logistic regression and Support vector machine. The generative models included Poisson regression from Dixon-Coles paper [7] and a bivariate Poisson regression that is an extension of the former one first introduced by [12]. The main purpose of this paper is to maximize the predictive performance of the generative Poisson model and compare it to other common methods in the industry, such as the ensembles, as a reference point. All the models were evaluated on their performance of the 1X2 prediction task and compared to the bookmakers' odds as well as a baseline model that predicts the most likely outcome based on the historical distribution of the target variable. The results show that all the models fall short of the bookmakers' odds, however, the best performing model was the bivariate Poisson regression with covariates, which was just marginally underperforming the market. This was the result of finding ways via algorithmic improvements as well as reducing the features space. Despite the other models trailing by a larger margin, they all very comfortably outperform the baseline model, which is a clear indication that the methods used in this paper are able to capture the signal in the data and learn patterns that are relevant for the prediction task. The results of this paper can be used as a reference point for future research in this area, as well as a benchmark for other models that are used in the industry.

2 Introduction

3 Methodology

To approach a problem, we firstly need to define it in rather strict terms. What we aim for in this paper is to predict the outcome of a football match. Meaning, we predict whether a football match is either going to be won by one or the other team or the result will be a draw, as is the common format in domestic football leagues where there is no overtime included, therefore a possibility for a draw. This kind of format is also called 1X2 in the betting industry. The 1 stands for home team win, X for a draw and 2 for the away team win. For this task, different kind of machine learning algorithms will be applied and evaluated based on their metrics. For training the ML models, it is of great importance to choose a dataset rich in information

and possible signal for the algorithms to learn and distinguish important patterns that may give the edge towards a correct prediction. The methodology of this paper could be defined in these steps:

- **Data exploration** Finding a dataset that is rich in information and contains the relevant features for the prediction task. Following that, the dataset is explored to find any patterns or distribution shifts that may occur over time. This includes looking at the distribution of the target variable, as well as the distribution of the features.
- **Feature engineering** Creating new features that may be relevant for the prediction task. This includes creating rolling averages of the features, as well as creating new features such as points per game, team elo ranking and expected goals (xG). This goes hand in hand with training the models and evaluating their performance, e.g some features do not serve any purpose and are rather creating noise than signal. Therefore this also includes feature selection practices, such as permutation importance and random forest feature importance.
- **Building the models** Model selection happens in the feature engineering phase already, where there is sort of a baseline created and everything that follows aims at one goal; to improve the performance of the models. The models are built and trained on the dataset, with the goal of maximizing their predictive performance. This includes hyperparameter tuning, as well as trying different architectures, especially going particularly deep in the generative models.
- **Train/test** Different ways of splitting the data are explored, such as training a single season and testing on the next one, training and testing within a single season or training multiple seasons and testing on the next one. There are certain limitations due to time series nature of the data. For some models it is also necessary to leave out a portion of data for validation purposes to avoid overfitting.
- **Upscaling** This includes the process of trying to boost the performance of models by adding more features which goes hand in hand with feature engineering. On top of exploring the feature space, it is also beneficial to simply add more seasons to the training data. In this paper the feature however come from a single league, meaning that adding data represents adding older seasons within the same league. This is a trade-off between adding more data and the distribution shift that occurs over time, as the game of football evolves and changes over time. A similar issue would be raised by adding data from other leagues, as the style of play is different and therefore the distribution of features would be different as well.
- **Evaluation** As well as other points in the methodology, evaluation is an ongoing process that is done throughout the majority of the project. The baseline models are evaluated on the simple and intuitive accuracy metric. Despite delivering the incomplete message, it has been proven in the results that accuracy is highly correlated with the more insightful measurements of performance on probabilistic outputs - log loss and Brier score. Therefore, it is deemed a sufficient metric for the baseline versions of models that are explored in this paper. However, as the models progressed along the feature space, log loss was the primary target for optimization.

4 Feature space

4.1 Dataset

The dataset curated from <https://www.football-data.co.uk/> represents the majority of the features used for machine learning. While the extracted dataset provided many features that could potentially be crucial in prediction of other match statistics, some of these proved to be of little value to our 1X2 prediction task. It is for this reason that additional features were added. Majority of them were derived from the existing dataset. Among the newly created feature was the **table position**. That was calculated for each team and each match, both the home and away team as well as the position difference between them. This gives the model an idea of the strength of the teams that are playing against each other. Next, from every existing feature we made sure there was a rolling average for home and away team as well as the difference between them. As shot accuracy was not found in the dataset, it was computed simply by dividing the number of goals scored

by total shots of the particular team. Since it was found in EDA, when transforming the features into these rolling averages and such, they were grouped not only by team but by the venue as well. This means that it was much stressed whether the team was the home team or away team, since the rolling averages of features were computed from its last 5 games from the venue they played at, shall it be home or away match.

4.2 Exploratory data analysis

The dataset available contains the last 12 seasons of the English Premier League (2014-2026), each season having 380 matches, that would add up to 4560 matches in total (disregarding the rare occasional incomplete entries that were dropped from the dataset). Distribution shift occurs in many aspects of the game when looking at football over a decade ago. Victoria Kereki in her paper about FC Barcelona's tactical Evolution (2004–2021) [13] talks about how clearly the game has changed when doing a clustering analysis on their passes throughout the years. In the span of 12 years of the EPL, there have also been noticeable changes in game patterns.

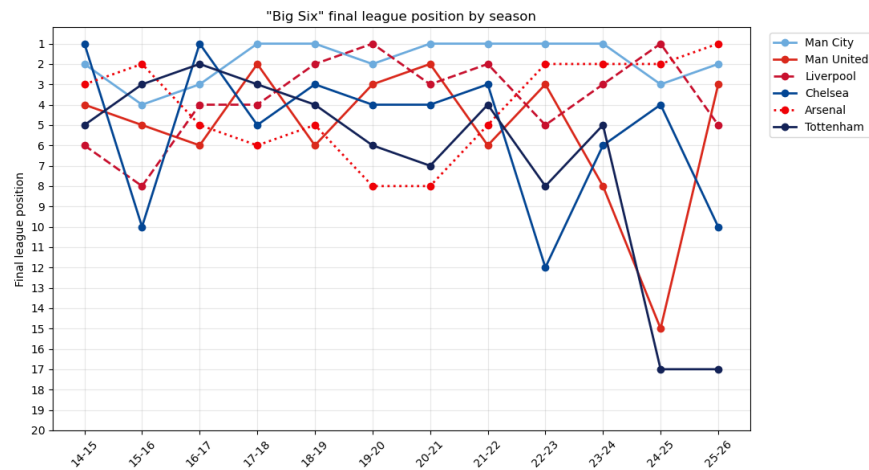


Figure 1: The "Big six"

Naturally, teams develop over time, whether into a good or bad direction. Figure 1 shows final standings in all 12 seasons of the so called "Big six" of the EPL teams. There is an obvious trend in recent years that shows a large variance in the final standings between most of these teams. This clearly indicates unpredictability. Moreover, the winner of 2015-16 EPL, Leicester, has in the 2025-26 season been relegated to League one - the **third** best league in England. Another evidence of a distribution shift can be seen on the figure below.

Home advantage has been very evident throughout all the seasons except one, when there were empty stadiums during the games due to Covid. This factor can negatively impact the predictions of a model, that naturally learns patterns about home advantage in prior seasons. The figure also shows the distribution of the target feature itself - the full time result of a game. Draws have always been the least likely outcome, even though often almost on equal with the away team wins.

Looking at the shots on target can also reveal some information about the trend over the last 12 years.

More shots on target results in more goals as well and in the EPL, they have been quite consistent throughout the years, except from the 2023-24 season, that is effectively an outlier, as the shots on target dropped rapidly in the following seasons again.

When plotting the overall count of goals per match in all the seasons together, an interesting pattern occurs.

The distribution of goals per match happens to follow the well-known Poisson distribution and it is tested by the χ^2 goodness-of-fit test on the total number of goals scored per match (home plus away), pooled across all 12 seasons in the dataset ($n = 4315$ matches).

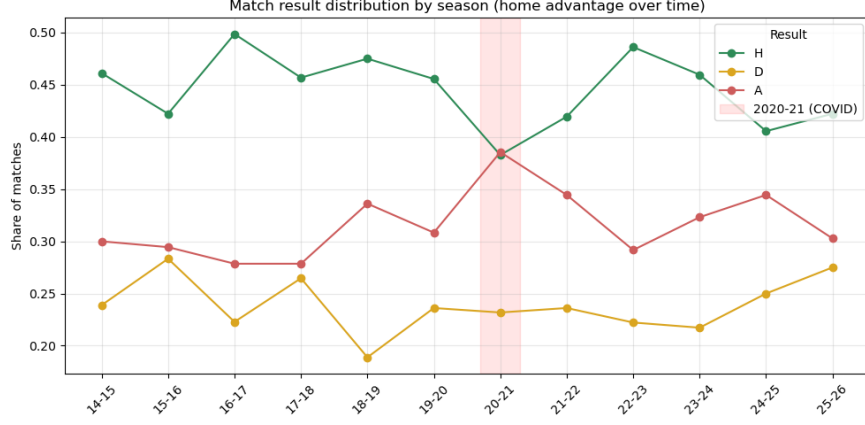


Figure 2: The home advantage distribution

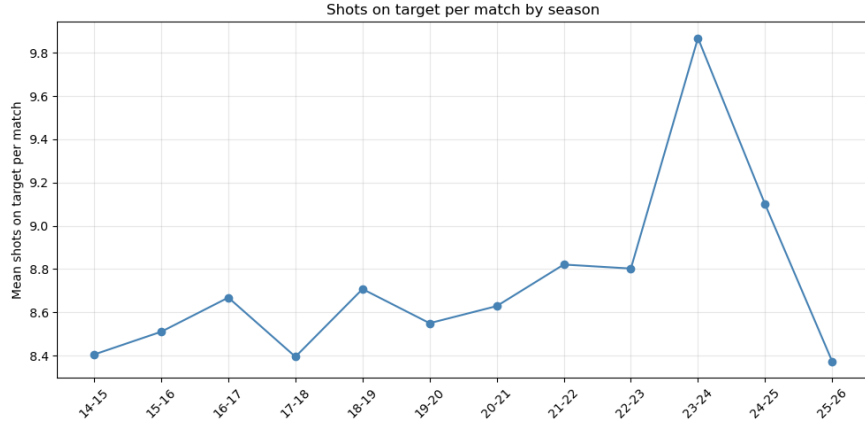


Figure 3: The shots on target distribution

The rate parameter $\hat{\lambda}$ was estimated via maximum likelihood as the sample mean:

$$\hat{\lambda} = \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i = 2.8019$$

with sample variance $s^2 = 2.7394$, close to $\hat{\lambda}$ as expected under the Poisson equidispersion property ($\text{Var}(X) = \mathbb{E}[X]$).

Under the null hypothesis H_0 : total goals per match $\sim \text{Poisson}(\hat{\lambda})$, the expected count in bin k is

$$E_k = n \cdot P(X = k) = n \cdot \frac{\hat{\lambda}^k e^{-\hat{\lambda}}}{k!}$$

for $k = 0, 1, \dots, 6$, with the tail bin ($k \geq 7$) pooled to keep expected counts sufficiently large:

$$E_{7+} = n \left(1 - \sum_{k=0}^6 P(X = k) \right)$$

Table 1 reports the observed (O_k) and expected (E_k) counts per bin. The Pearson χ^2 test statistic is

$$\chi^2 = \sum_k \frac{(O_k - E_k)^2}{E_k} = 9.2502$$

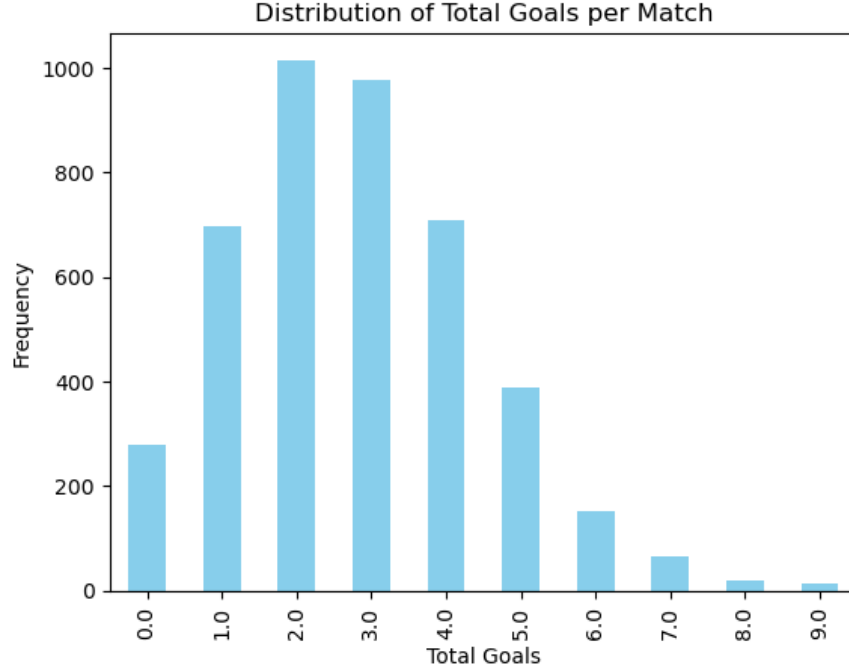


Figure 4: Distribution of total goals per match

Table 1: Observed vs. expected total-goals-per-match counts under $\text{Poisson}(\hat{\lambda} = 2.802)$

Goals	Observed O_k	Expected E_k
0	279	261.91
1	696	733.83
2	1015	1028.04
3	975	960.14
4	709	672.55
5	389	376.87
6	153	175.99
7+	99	105.66

with $\text{dof} = (\text{bins} - 1) - (\text{estimated parameters}) = 8 - 1 - 1 = 6$ (one degree of freedom is lost for estimating $\hat{\lambda}$ from the data). This yields

$$p = 0.1600$$

Since $p > 0.05$, the null hypothesis is not rejected: the observed distribution of total goals per match does not deviate significantly from a Poisson distribution. This supports the use of Poisson (and bivariate Poisson) regression as an appropriate generative model for goal counts in this project.

4.3 Feature importance

To avoid any ambiguity in the choice of features that would pass through feature selection, a simple random forest model was run on the dataset, using the features that carry the information that is readily available before the match, not during it, therefore rendering the `Half-time result` and `Half-time home and away goals` features unusable. After the inference, the `feature_importances_` method from `sklearn` determined the features' importance based on impurity of the leafs in the decision trees. The feature importance function revealed the features that this random forest model was making it's decisions based on. That is one way to

Feature	Importance
AST (away shots on target)	0.2784
HST (home shots on target)	0.2612
AS (away shots)	0.1213
HS (home shots)	0.0930
AF (away fouls)	0.0444
AC (away corners)	0.0418
AY (away yellows)	0.0390
HF (home fouls)	0.0388
HC (home corners)	0.0376
HY (home yellows)	0.0313
HR (home reds)	0.0104
AR (away reds)	0.0027

Table 2: Feature importance ranking

address the issue of feature selection. On this basis, the shot features, especially the ones on target, were dominant. On the other hand, red card features rank very poorly, in the 0.01-0.002 territory. The final decision was to cut out the red card features as they were by far the smallest, negligible contributor to the model.

The current state of the feature space offers a baseline selection of predictors for the match outcome. However, despite these features representing the essential statistics of a football game, the accuracy of the model could be improved with more advanced predictors in the dataset. That is why it is necessary to look further and take the feature selection to another level.

4.4 Points per game

Another way of capturing the strength of a team are their respective points per game. This directly encapsulates team strength not just in terms of their shots or fouls they make, but the points they gain over the course of the season. This is naturally a number between 0 to 3.

4.5 Team elo ranking

One of the extra features that do not occur in the original dataset is called Elo. Elo ranking was first introduced for determining the relative skill level of chess players. [16] In the case of football statistics, it is an adaptation of the original formula, with a twist of considering the magnitude of win, namely the goal difference, rather than a simple win/loss scenario that occurs in chess.

The expected outcome for the home team is given by:

$$E_H = \frac{1}{1 + 10^{-(R_H - R_A)/400}} \quad (1)$$

E_H Expected score (probability of winning) for the home team, $E_H \in [0, 1]$.

R_H Current Elo rating of the home team.

R_A Current Elo rating of the away team.

The corresponding rating update after the match is played is:

$$R'_H = R_H + K \cdot G \cdot (S_H - E_H) \quad (2)$$

R'_H Updated Elo rating for the home team after the match.

K Development constant controlling the maximum possible rating adjustment per match.

G Goal-difference multiplier, scaling the rating change according to the margin of victory (commonly used in football Elo variants, e.g., the World Football Elo Ratings system).

S_H Actual match outcome for the home team: $S_H = 1$ for a win, $S_H = 0.5$ for a draw, $S_H = 0$ for a loss.

E_H Expected outcome for the home team, as defined above.

A common specification for the goal-difference multiplier G , used by the World Football Elo Ratings, is:

$$G = \begin{cases} 1 & \text{if goal difference} = 0 \text{ or } 1 \\ 1.5 & \text{if goal difference} = 2 \\ \frac{11 + |\text{goal difference}|}{8} & \text{if goal difference} \geq 3 \end{cases} \quad (3)$$

[8] Since Elo rating is already capturing a lot of the information about wins and losses and the goals scored while being relative to each team's strength, it is considered to be the cornerstone of the feature dataset. Elo ratings are obtained via the <http://www.clubelo.com> API. Filtering based on team's name, every single match in the dataset contains the current team's elo rating at a time. This elo rating is being updated consequently following the formula explained earlier in this section so that it encapsulates the long term strength of a team but carries information about the recent form as well.

4.6 Moving average

First and foremost, when predicting a match result before the data is available, it is necessary to have some data to feed the model beforehand. This can be provided by the previous matches of both teams. Therefore, this problem is handled by computing the moving averages of the features used in the dataset. The size of the window chosen was based on running a sweep over a range of configurations. Instead of a plain rolling average, Exponentially weighted moving average (EWMA) was used.

$$\text{EWMA}(t) = \alpha \cdot x(t) + (1 - \alpha) \cdot \text{EWMA}(t - 1) \quad (4)$$

- $x(t)$ is the current data point.
- α is the smoothing factor, value between 0 and 1. The higher the value the more weight added to recent data
- $\text{EWMA}(t - 1)$ is the previous EWMA value.

Intuitively, the recent form for sports such as football means last few matches, however, with the feature engineering loop through the moving average window, it has been proven that the convergence of model's log loss was reached at number 14. This is a number that represents over half of a single season, when it comes to the features that are grouped by venue for a team. To be precise, since each team plays 38 matches in total, split equally in home and away venues, the moving average window spans across $14/19 = 74\%$ of seasonal matches, for the venue specific features. This might seem like a lot, but that is exactly why the EMWA is such a good fit. Ability to capture smoothed signal from the data while prioritizing the most recent matches is what separates this method from a vanilla rolling average. For reference, the smoothing factor is using the standard **Pandas** formula that scales the smoothing factor according to the span size; $\alpha = 2 / (\text{span} + 1)$. Half-life (time for a weight to decay to 50%): solve $(1 - \alpha)^h = 0.5$ for h . With $\alpha = 2/15$, $1 - \alpha = 13/15 \approx 0.8667$:

$$\alpha = \frac{2}{\text{span} + 1} = \frac{2}{14 + 1} = \frac{2}{15} \approx 0.1333 \quad (5)$$

$$w_k = \alpha(1 - \alpha)^k \quad (6)$$

$$h = \frac{\ln(0.5)}{\ln(1 - \alpha)} = \frac{\ln(0.5)}{\ln\left(\frac{13}{15}\right)} \approx \frac{-0.6931}{-0.1431} \approx 4.84 \text{ matches} \quad (7)$$

Contribution of the match at the span boundary (14 matches back, relative to the most recent match’s weight): the weight of an observation k lags back is $w_k = \alpha(1 - \alpha)^k$, so the ratio to the most recent ($k = 0$) is just $(1 - \alpha)^k$.

$$\frac{w_{14}}{w_0} = (1 - \alpha)^{14} = \left(\frac{13}{15}\right)^{14} \approx 0.1349 \text{ (13.49\%)} \quad (8)$$

For $k = 14$: $(13/15)^{14} \approx 0.1349 \rightarrow$ the match 14 games back contributes 13.5% as much as the most recent one.

4.7 Expected goals (xG)

Another feature that is a great contributor to the explanation of the variance in data is the xG statistic. An expected goals model tries to estimate the probability of any given shot being converted to a goal based on various different factors describing the shot. These probabilities can then be added up per team and yield a “result-agnostic” description of the teams’ performance.[1] The xG feature alone is estimated by a supervised machine learning model that outputs a probabilistic number between 0 and 1. Whichever discriminative method used, features such as shot location, speed of player taking the shot, defenders in the line of the shot or goalkeepers position are pivotal in the xG estimation.

The xG values were obtained from Understat.com, a third-party analytics provider. It is important to note that Understat’s exact xG model is proprietary and not published in peer-reviewed literature; the values used here should be treated as an externally-validated but non-transparent measure of chance quality.

Rank	Feature	Importance
1	Elo_Difference	0.0810
2	PPG_Difference	0.0436
3	Home_ShotOnTargetDifference_Rolling	0.0352
4	Home_ShotOnTargetDifference_RollingTeam	0.0252
5	Home_Elo	0.0220
6	Home_ShotDifference_Rolling	0.0217
7	Home_deep_Rolling	0.0215
8	Home_ShotsOnTarget_Rolling	0.0213
9	Home_Shots_Rolling	0.0203
10	Away_ShotOnTargetDifference_RollingTeam	0.0196
11	Away_Elo	0.0194
12	Away_xG_RollingTeam	0.0194
13	Away_ShotDifference_RollingTeam	0.0193
14	Home_xG_Rolling	0.0191
15	Home_GoalDifference_Rolling	0.0177

Table 3: RF Feature importance ranking after increasing the feature space

However, when moving away from the random forest’s gini impurity based feature importance to the permutation importance, we see a bit different set of top 15 features in the ranking.

5 Bookmakers

Odds for sports events, most of all football odds, have been out there for a long time. There is a whole industry that is based on creating these and their entire business model relies on how precise and close to the ground truth their predictions are. As stated by the American gaming association, the sports betting industry hit a record revenue figure of \$16.96 billion in 2025[14]. This number indicates that despite the difficulty of predicting outcomes in sports where there could be hundreds or thousand of deciding factors, there are still billions of dollars of profit the US alone by the betting companies. These companies or people

Rank	Feature	Mean Δ log loss
1	Home_ShotOnTargetDifference_Rolling	+0.0016
2	Home_ShotOnTargetDifference_RollingTeam	+0.0013
3	PPG_Difference	+0.0013
4	Home_Shots_Rolling	+0.0012
5	Elo_Difference	+0.0010
6	Home_ShotDifference_RollingTeam	+0.0010
7	Home_ShotsOnTarget_Rolling	+0.0009
8	Home_xGA_Rolling	+0.0008
9	Home_ShotDifference_Rolling	+0.0008
10	Home_deep_Rolling	+0.0008
11	Home_GoalDifference_RollingTeam	+0.0005
12	Away_PPG	+0.0004
13	Home_GoalDifference_Rolling	+0.0004
14	Home_xGA_RollingTeam	+0.0004
15	Away_ShotAccuracy_Rolling	+0.0003

Table 4: Permutation feature importance ranked by mean Δ log loss

are often referred to as "Bookmakers". However, even with the state of the art machine learning models on top of carefully curated and engineered data, there is still a very important factor that is much responsible for the huge profit of bookmakers. Similar to casinos, bookmakers do not play "fair". What I am referring to now is called the margin. That is a small percentage of the value that is taken away from the odds that are on the market, regardless of the particular bookmaker. To explain this in detail, I must first address how odds work. There are a few different ways that one can interpret odds, such as fractional (**3/2**), American (**+150**) or decimal (**2.5**) which I will be using in this paper. The decimal odds are given by **1/probability**. In the 1x2 betting, there are three options of the final outcome of a game. Therefore, the probabilities of these three options together must sum to 1. To demonstrate on an example of a 40% probability for home team win, 25% probability for a draw and 35% probability for away team win:

$$\begin{aligned}
\text{Home team: } \frac{1}{0.40} &= 2.5 \\
\text{Draw: } \frac{1}{0.25} &= 4 \\
\text{Away team: } \frac{1}{0.35} &= 2.86
\end{aligned}$$

These odds reflect the true probability, or rather the assumption of it. What I found in the [10] dataset is that when all the odds were added together after each dividing 1; $(1/H + 1/D + 1/A)$, the sum of them was not 1 but somewhere around 1.048 on average across all the matches in the dataset. That is precisely what the margin represents. This is naturally in favour of the bookies, because when . It has also been found that the 4.8% margin is spread across all three outcomes of a match fairly evenly. The relative gaps between

Outcome	Implied	Actual	Gap (pts)	Gap relative to its own probability
Home	46.4%	44.8%	+1.61	3.5%
Draw	25.7%	23.9%	+1.75	6.8%
Away	32.8%	31.3%	+1.47	4.5%

Table 5: Bookmaker odds calibration: implied vs. actual outcome probabilities

the implied and actual probabilities are however different, indicating that when not sure, bookmakers add more margin to the probability of an event happening, therefore lowering the odds for such event by a bigger relative amount. This is most commonly found in determining the draw odds.

6 Models

As the title of this paper is hinting at, the goal is to see which machine learning models do well in the task of predicting a football match result. There is a variety of ML models that work on a large spectrum of methods that ultimately yield the same thing - probabilistic prediction. That is why it might be interesting to look at whether there are any major differences in terms of which some algorithms might be favoured for this particular prediction problem rather than others. Before getting into models' specifics, it is worth mentioning again the shape and size of the training data. Football matches records are in tabular form that contain roughly 1000 entries that would primarily contain the information from previous matches of each team and a number (elo) that describes team's strength. This data is also time-sensitive, meaning that not following the continuity of matches in time would result in invalid machine learning practices. This is especially important in partitioning the data into train/validation/test splits that also go hand-in-hand with the cross-validation.

6.1 Splitting the data

Due to the distribution shift in many aspects of the game shown in the EDA section of this paper, it is unfeasible to train a model on matches that were carried out over 10 years ago and test it on the current season. This can be caused by transfer of basically all the players, changes of head coaches and team staff, internal team factors as well as other external factors that can occur over such a long period. Nevertheless, this can be handled by training over a smaller window of seasons. However, since there are 12 seasons available, what is implemented in the train/test split here is called a walk-forward validation method. This method serves as cross-validation. It is implemented because of the need for time-series relevant division of train/test data. It utilizes a rolling window approach where the model is trained and tested on consecutive periods. This technique allows the model to be updated continuously, closely simulating real-world forecasting scenarios while maintaining temporal order. [9] It has a consistent window size and can detect concept drift in terms of the variable being predicted.

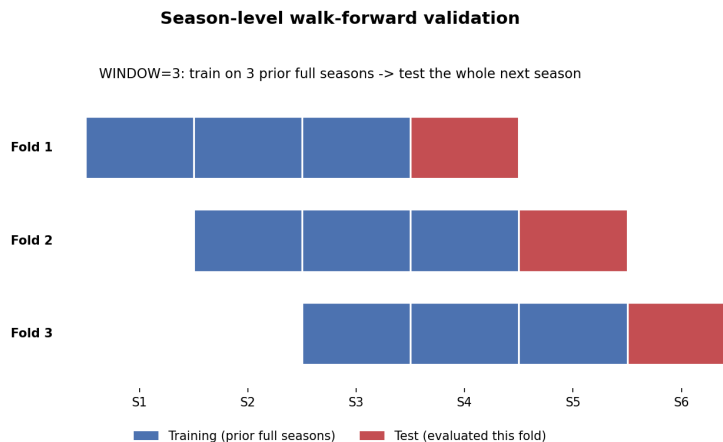


Figure 5: Walk-forward scheme

Walk-forward method is used for two reasons. Firstly, it is necessary to perform some sort of cross-validation in order to reduce randomness and providing robust reliable results. Secondly, as mentioned earlier, reducing the size of training is in this case preferred, as going back in time too far can be counterproductive. The parameter tuning, in this case the parameter being the window size of the walk-forward validation, has revealed that the ideal train set would include 3-4 seasons. On top of the walk-forward validation algorithm, the training consists of within-season chunks that include the freshest set of matches for making predictions.

The chunks accumulate over time and for each window of the span of 3 seasons. This means that for each window size, e.g iteration of the walk-forward mechanism, there are actually 5 iterations for each window. There are 9 windows, since there are 12 seasons in total. That also means that the first three seasons from

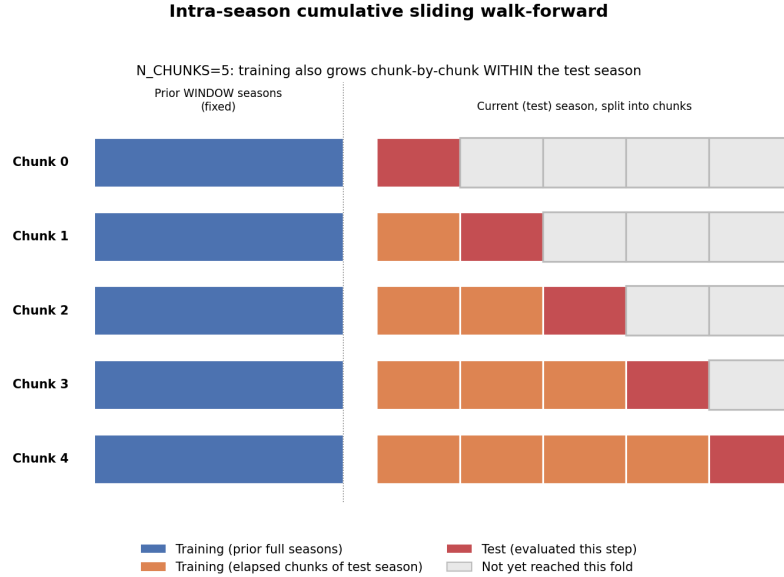


Figure 6: Walk-forward intraseason scheme

2014-2016/17 are excluded from testing. In total, there are 9 windows multiplied by 5 cumulative chunks for each, making it 45 iterations of the model. Validation set is not incorporated in this scheme, as there is no need for it since this is the final evaluation method of the models that have already been tuned and tested for overfitting.

6.2 Neural network

First method on the list is a regular feed-forward neural network, utilizing the **neural-network** family from **sklearn**. This is a simple implementation that the package offers.

Listing 1: FFNN configuration

```

1 nn = MLPClassifier(
2     hidden_layer_sizes=(8,8),
3     activation="relu",
4     solver="adam",
5     alpha=5.55,
6     learning_rate_init=0.001,
7     random_state=42
8 )

```

The fine-tuning was ran on two hyperparameters. First of them, the hidden layers. Because the data that the model is being fed is of small size relative to the neural nets standard, the selection of the hidden layers that were chosen for fine-tuning was also chosen as for a small model. [(8,), (16,), (32,), (8, 8), (16, 8)]. The second variable that went through the fine-tuning sweep was the ridge regression (L2) regularization term referred to as "alpha" in the **sklearn** implementation. It penalizes the sum of squared coefficients and shrinks the weights close to zero. The default value of alpha that this package uses is 0.0001, so the 5.55 value that is actually used is 55 500 times bigger. This points at the fact that the dataset is substantially smaller than what would typically be used for neural networks. Therefore, the data which already includes high variance must be strongly regularized and the fine-tuning discovered the unusually large L2 term. Using a smaller would result in the model seeing and memorizing patterns in noise.

6.3 Logistic regression

A classic machine learning method that models the log-odds based on a linear combination of features. Then uses the softmax function to compute the probabilities of either of the three events (H/D/A) happening. The data is first scaled by `sklearn`'s `StandardScaler` class that centers the data around its mean and scales it based on the variance with the formula

$$z = \frac{x - \mu}{\sigma} \quad (9)$$

. Other than that, the configuration of the predictor was as such:

Listing 2: Logistic regression configuration

```
1 lr = LogisticRegression(max_iter=2000, solver="saga", random_state=42, "C": 0.001,
2   "penalty": "l2")
```

A minor fine-tuning sweep was conducted on the logistic regression as well, choosing the regularization factor L2 over L1 and the "C" which in this case is also a regularization factor, similar as to the one in FFNN, except that the smaller it is the more regularization, thus higher the penalty term. The variable C just describes the inverse of the strength of the penalty term, be it L1 or L2. The intuition follows the same pattern as in the previous model, and that is the high per-match variance in football requires large regularization terms as to prevent learning on noise and therefore overfitting.

6.4 Poisson regression

Unlike the rest of the discriminative models in this paper, the Poisson regression is a generative one. It is a statistical technique used to model and analyze count data, where the outcome variable represents the number of times an event occurs in a fixed interval of time, space, or any other dimension. It is most appropriate when the values of the response variable are non-negative whole numbers. [11] The key assumption is that the counts follow a Poisson distribution. It has been proven in the EDA section of this paper that the goals scored per match do indeed follow this distribution. That is why Poisson regression can be used in this scenario, specifically to model the number of goals for each team, therefore determining the final match outcome as well.

6.4.1 Dixon-Coles (1997) Poisson goal model [7]

Each team gets an attack strength and a defense weakness ratio. Both home and away goals are modeled as two Poisson-distributed counts. Home advantage gets also factored in as another ratio term. The assumption about the double Poisson model is that these two distributions are independent, except for one case. A correlation correction ρ is added to address the issue of independent Poisson underestimating the amount of low-scoring matches that actually occur where there are 0-2 goals in total, therefore 4 of the plenty of possible outcomes (0-0, 0-1, 1-0, 1-1). Dixon and Coles motivate this dependence tactically: in a tight, low-scoring game both sides have an incentive to shift toward more cautious, risk-averse play in order to protect the current scoreline, which suppresses further goals for *both* teams simultaneously - a joint effect that two independent Poisson variables cannot represent, and one that fades once the game is no longer close. This model also introduced the exponential time-weighting decay, which is essentially the same mechanism that is used in the exponential moving weighted average in the feature set. However, this model only ever consumes HomeTeam/AwayTeam/FTHG/FTAG features - team names and their goal counts.

Parameters that are needed for making a prediction, α - attack strength, β - defense strength and γ - home advantage are estimated via Maximum likelihood estimation, as there is no closed formula solution. The optimizer chosen was the BFGS algorithm from `scipy.optimize.minimize` package, well suited for differentiable functions without constraints. The likelihood function for the Dixon-Coles model is defined as follows:

$$\mathcal{L}(\alpha, \beta, \gamma, \rho) = \prod_{k=1}^N \tau_{\rho}(x_k, y_k) \cdot \frac{e^{-\lambda_k} \lambda_k^{x_k}}{x_k!} \cdot \frac{e^{-\mu_k} \mu_k^{y_k}}{y_k!} \quad (10)$$

N Total number of matches in the dataset.

x_k Actual number of goals scored by the home team in match k .

y_k Actual number of goals scored by the away team in match k .

λ_k Expected home goals in match k , as defined by $\log(\lambda_k) = \alpha_{i(k)} + \beta_{j(k)} + \gamma$.

μ_k Expected away goals in match k , as defined by $\log(\mu_k) = \alpha_{j(k)} + \beta_{i(k)}$.

$\tau_\rho(x, y)$ Correction factor applied to the independent Poisson probability for scoreline (x, y) . This is what separates the Dixon-Coles model from just a plain double Poisson regression.

ρ Dependence parameter, estimated jointly with α, β, γ via MLE, typically found to be small and negative (around -0.1 to -0.2 in most fitted football datasets).

The correction factor τ is defined as follows:

$$\tau_\rho(x, y) = \begin{cases} 1 - \lambda\mu\rho & \text{if } x = 0, y = 0 \\ 1 + \lambda\rho & \text{if } x = 0, y = 1 \\ 1 + \mu\rho & \text{if } x = 1, y = 0 \\ 1 - \rho & \text{if } x = 1, y = 1 \\ 1 & \text{otherwise} \end{cases} \quad (11)$$

A team with no history in the training window (e.g. newly promoted) falls back to average league strength, $\alpha_i = \beta_i = 0$.

After estimating the coefficients on the particular training set, coefficients like shown on an example here are ready for the next stage. The higher the attacking coefficient, the more likely is a team to score more goals, while the lower the defense, the harder it should be for the opposition to score.

Team	Attack	Defense
Arsenal	1.034	-0.939
Aston Villa	0.843	-0.689
Bournemouth	0.812	-0.652
Brighton	0.751	-0.832
Burnley	0.874	-0.917
Chelsea	1.241	-0.870

Table 6: Example of team attack and defense strength ratings

The prediction formula using the coefficients from previous step:

$$\log(\lambda_{ij}) = \alpha_i + \beta_j + \gamma \quad (12)$$

$$\log(\mu_{ji}) = \alpha_j + \beta_i \quad (13)$$

λ_{ij} Expected number of goals scored by the home team i against away team j .

μ_{ji} Expected number of goals scored by the away team j against home team i .

α_i Attack strength coefficient of team i .

β_j Defense strength coefficient of team j .

γ Home advantage coefficient.

6.4.2 Dixon-Coles Poisson model with Covariates

The distinction between the classic Dixon-Coles Poisson model described above and the extension of it with covariates is in how λ and μ (each team's expected goals) get computed.

$$\log(\lambda) = \text{home_adv} + \beta_{\text{home}} \cdot [\text{covariates}] \quad (14)$$

$$\log(\mu) = \beta_{\text{away}} \cdot [\text{covariates}] \quad (15)$$

Perhaps contrary to intuition, adding more features in this model, unlike only using team name and team goals features in the first model actually produces less parameters. The parameters are shrunk from $2(\text{home and away team}) \cdot 20(\text{number of teams}) + (\text{home advantage} + \rho) = 42$ total to $16(\text{home advantage} + \beta + 2 \cdot 7(\text{covariates for home and away team}))$. Team identity is never used directly here; the model only ever sees each team's current features without explicitly assigning a team name to it.

All the 16 parameters are estimated via Maximum likelihood estimation. The log-likelihood function is the same as in the simple Dixon-Coles model, except that λ and μ are computed differently. The initialization of the parameters is as follows: $\beta = 0$, $\text{home_adv} = 0.2$, $\rho = 0$. The covariates variable in the equations represents a vector of features that are used to predict the expected goals for each team. The coefficients β_{home} and β_{away} are estimated for these covariates, allowing the model to capture the influence of various factors on the expected goals scored by each team. The covariates undergo standardization before the optimization takes place, as the coefficients are estimated on a standardized scale. Covariates were selected carefully after running a feature importance analysis in the earlier section of this paper. The final set of covariates that yielded best results were:

Listing 3: Selected covariates for the Dixon-Coles covariate model

```

1 HOME_COVARIATES = [
2     "Home_Elo", "Away_Elo", "Home_xG_Rolling5", "Away_xGA_Rolling5", "Home_PPG",
3     "Home_ShotOnTargetDifference_RollingTeam7", "
4     Away_ShotOnTargetDifference_RollingTeam7",
5 ]
6 AWAY_COVARIATES = [
7     "Away_Elo", "Home_Elo", "Away_xG_Rolling5", "Home_xGA_Rolling5", "Away_PPG",
8     "Away_ShotOnTargetDifference_RollingTeam7", "
9     Home_ShotOnTargetDifference_RollingTeam7",
10 ]

```

6.4.3 Bivariate Poisson model with Covariates

As an extension to the Dixon-Coles model, the bivariate Poisson model with covariates allows for modeling the joint distribution of two count variables (e.g., goals scored by home and away teams) while incorporating covariates. This model captures the potential correlation between the two count variables, which is particularly relevant in football matches where the performance of one team can influence the performance of the other. This is the exact opposite assumption of the key assumption in the Dixon-Coles model, where the two Poisson distributions are assumed to be independent, except for the low-scoring matches. The bivariate Poisson model with covariates allows for a more flexible representation of the relationship between the two count variables, enabling a better and far more realistic understanding of how various factors impact the expected goals for each team [12]. This method is therefore a more general approach to modeling football match outcomes as opposed to a hand-patch fix for the dealing with dependence between two variables in low-scoring matches addressed in the Dixon-Coles model.

Instead of two independent Poisson counts, define three independent Poisson variables X_1, X_2, X_3 and set:

$$\text{Home} = X_1 + X_3, \quad \text{Away} = X_2 + X_3$$

X_3 is a goals component shared by both teams (interpretable as game pace/tempo), which induces $\text{Cov}(\text{Home}, \text{Away}) = \lambda_3$. $\log \lambda_1$ (home rate) and $\log \lambda_2$ (away rate, previously denoted as μ) are driven by the exact same 7 standardized covariates as your independent-covariate model. $\lambda_3 = e^\theta$ is a single global scalar that takes over the

role of ρ as the model's one correlation parameter. The parameter count remains unchanged at 16 (β_{home} (7) + β_{away} (7) + home_adv + θ), all still fit jointly via the same BFGS MLE as before.

$$\mathcal{L}(\beta_{\text{home}}, \beta_{\text{away}}, \gamma, \theta) = \prod_{k=1}^N e^{-(\lambda_{1,k} + \lambda_{2,k} + \lambda_3)} \frac{\lambda_{1,k}^{x_k}}{x_k!} \frac{\lambda_{2,k}^{y_k}}{y_k!} \sum_{i=0}^{\min(x_k, y_k)} \binom{x_k}{i} \binom{y_k}{i} i! \left(\frac{\lambda_3}{\lambda_{1,k} \lambda_{2,k}} \right)^i \quad (16)$$

with

$$\lambda_{1,k} = \exp!(\gamma + \mathbf{c}_{\text{home},k} \cdot \beta_{\text{home}}), \quad \lambda_{2,k} = \exp!(\mathbf{c}_{\text{away},k} \cdot \beta_{\text{away}}), \quad \lambda_3 = e^\theta \quad (17)$$

N Total number of matches in the dataset.

x_k Actual number of goals scored by the home team in match k .

y_k Actual number of goals scored by the away team in match k .

$\mathbf{c}_{\text{home},k}, \mathbf{c}_{\text{away},k}$ Standardized covariate vectors for match k (Elo, rolling xG/xGA, PPG, shot-on-target difference).

$\lambda_{1,k}, \lambda_{2,k}$ Match-specific home/away scoring rates, driven by the covariates as in the independent covariate model.

λ_3 Shared goals component (interpretable as game pace/tempo) inducing $\text{Cov}(X, Y) = \lambda_3$ across every scoreline. Replaces ρ from the Dixon-Coles τ correction.

θ Free parameter estimated jointly with $\beta_{\text{home}}, \beta_{\text{away}}, \gamma$ via MLE, with $\lambda_3 = e^\theta$ enforcing $\lambda_3 > 0$ (only non-negative correlation is representable by this construction).

The final prediction of the match outcome is based on:

$$P(\text{H}) = \sum_{x=1}^K \sum_{y=0}^{x-1} P(X=x, Y=y), \quad P(\text{D}) = \sum_{x=0}^K P(X=x, Y=x), \quad P(\text{A}) = \sum_{y=1}^K \sum_{x=0}^{y-1} P(X=x, Y=y)$$

where K is the goal-truncation cutoff (max set to 10 in the code) and the grid $P(X=x, Y=y)_{x,y=0}^K$ is renormalized to sum to 1 before these three sums are taken.

6.5 Ensemble

Machine learning ensembles are a powerful machine learning method utilizing the wisdom of the crowd. The idea is to combine multiple models to produce a more accurate and robust prediction. Two of the popular methods are bagging and boosting. The main representatives of each are Random Forest [2] and XGBoost [5], respectively, tree based methods that are widely used in many machine learning tasks. These methods are particularly useful in tabular data which happens to be the format of the dataset in this paper. The ensemble methods are also known for their ability to handle high-dimensional data and capture complex relationships between features, making them suitable for the task of predicting football match outcomes.

6.5.1 Random forest

6.5.2 XGBoost

6.6 SVM

6.6.1 Simple SVM

6.6.2 SVM with UMAP

continuous feature importance

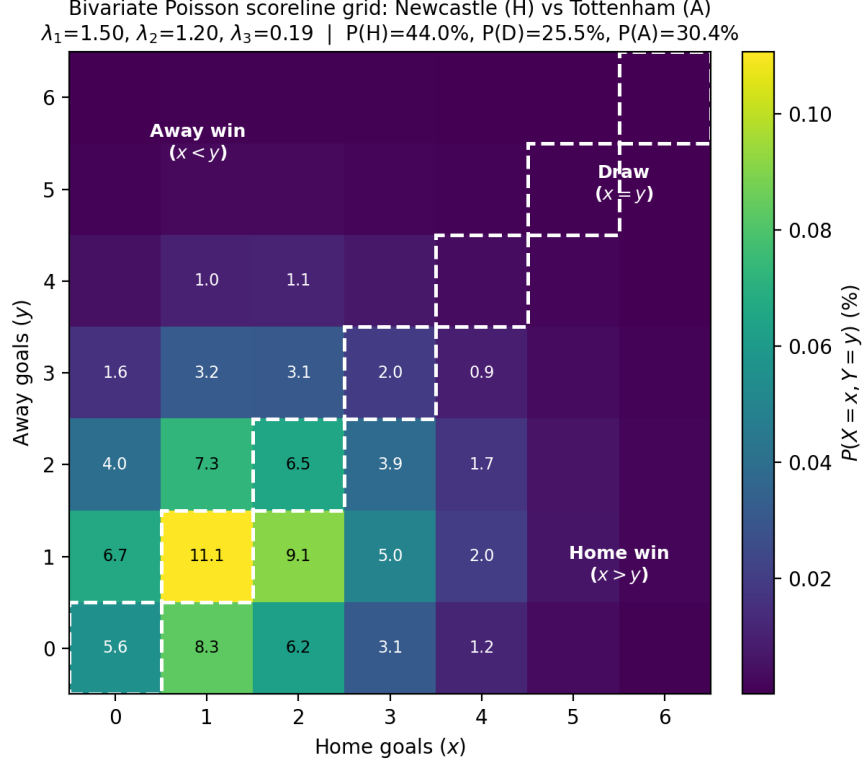


Figure 7: Example scoreline grid $P(X=x, Y=y)$ from the fitted bivariate Poisson model, for one real match (Newcastle vs. Tottenham, 2024-25 season, $K = 6$). The dashed cells mark the draw diagonal; summing the grid below/on/above it gives $P(H)$, $P(D)$, $P(A)$ respectively.

7 Evaluation methods

The ultimate goal of the models in 1X2 betting is to produce probabilistic output. In this paper, predictions are firstly treated as a classification problem. The model simply predicts the outcome of a match. This is already a great and interpretable way in order to carry out feature engineering, which is pivotal to increasing the accuracy and F1 score. However, accuracy, defined as the proportion of correctly predicted outcomes, is widely used but insufficient for probabilistic outputs[4]. This is due to the need for a way to measure the quality of probabilistic predictions. When a correct prediction is made with 90% confidence, it means that the model is very certain about its prediction. However, a model can predict the same outcome with a 51% confidence. The accuracy would not capture the uncertainty in the model's predictions, and that is the very essential task in the odds production. What is a much more insightful metric for evaluating probabilistic predictions is the logarithmic loss (log loss), which measures the performance of a classification model where the prediction input is a probability value between 0 and 1. Log loss increases as the predicted probability diverges from the actual label. A perfect model would have a log loss of 0. The log loss is defined as follows:

$$\text{LogLoss} = -\frac{1}{N} \sum_{t=1}^N \sum_{i=1}^R o_{ti} \log(f_{ti})$$

Equivalently, since $o_{ti} = 1$ only for the realized outcome and 0 otherwise, this collapses to just the log-probability the model assigned to what actually happened:

$$\text{LogLoss} = -\frac{1}{N} \sum_{t=1}^N \log(f_{t,y_t})$$

Another way of measuring the quality of probabilistic predictions is the Brier score. It is a strictly proper scoring rule that measures the accuracy of probabilistic predictions. The Brier score, similarly to log loss, is applicable to tasks in which predictions must assign probabilities to a set of mutually exclusive discrete outcomes or classes. [15].

The multi-category (multinomial) Brier Score generalizes the binary case to a set of mutually exclusive discrete outcomes or classes [15].

$$BS = \frac{1}{N} \sum_{t=1}^N \sum_{i=1}^R (f_{ti} - o_{ti})^2 \quad (18)$$

N Total number of forecast instances (e.g., matches).

R Total number of mutually exclusive outcome categories (e.g., $R = 3$ for home win, draw, away win).

f_{ti} Forecast probability assigned to category i for instance t , where $\sum_{i=1}^R f_{ti} = 1$.

o_{ti} Binary indicator for the actual outcome: $o_{ti} = 1$ if category i occurred for instance t , and $o_{ti} = 0$ otherwise.

It is therefore a much preferred method to evaluating probabilistic outcomes than accuracy, which disregards the magnitude of hits and misses completely. The Brier loss evaluation method comes at a cost of low interpretability, compared to raw accuracy. However, first introduced in weather forecast, the Brier skill score can handle this as well. The Brier Skill Score (BSS) expresses the relative improvement of a forecast model over a reference forecast, such as climatology (long-run base rates) or a naive baseline. [6]

$$BSS = 1 - \frac{BS_{\text{forecast}}}{BS_{\text{reference}}} \quad (19)$$

BS_{forecast} Brier Score of the model being evaluated.

$BS_{\text{reference}}$ Brier Score of the reference (baseline) forecast.

BSS Skill score, where $BSS > 0$ indicates the forecast model outperforms the reference, $BSS = 0$ indicates no improvement, and $BSS < 0$ indicates the model performs worse than the reference.

In the case of this paper, the reference that is being compared to the models' predictions are two numbers. One is the naive model that predicts classes randomly at a rate of 33.3% for each. The other one is a concept called climatological probability. First introduced in 1950[3], it works on the same principle as the naive model, except that it has knowledge of the real distribution of classes (H/D/A). After computing the BSS of both the models and the market odds, comparison can be made between these two scores.

8 Results and discussion

Log loss and the Brier score (BS) metrics are averaged across the 45 iterations of the walk-forward validation. The accuracy is computed as the total number of correct predictions divided by the total number of matches in the test set. The p-value is calculated using a paired t-test comparing the log loss of each model against the log loss of BET365's odds, which serves as a benchmark for comparison. The Brier Skill Score (BSS) is computed relative to the climatological probability baseline, which has been addressed in the evaluation section.

The key metrics to evaluating probabilistic models are the log loss and the Brier score even though the accuracy follows the trend of the other two metrics and is reported here for completeness. The table shows the distinction between discriminative and generative models. The discriminative models, which include logistic regression, feed-forward neural networks, random forests, XGBoost, and SVMs, generally perform worse than the generative Poisson models in terms of log loss and Brier score. Among the Poisson models, the bivariate Poisson model with covariates achieves the best performance, indicating that accounting for

Table 7: Model evaluation results (intra-season walk-forward validation, pooled over 3236 matches)

Model	Log Loss	BS	Accuracy	<i>p</i> -value vs. BET365	BSS
Logistic Regression	0.9752	0.5795	53.92%	0.000006	10.24%
FFNN	1.0002	0.5950	51.92%	< 0.000001	7.84%
Random Forest	0.9835	0.5845	52.53%	< 0.000001	9.47%
XGBoost	0.9923	0.5903	52.32%	< 0.000001	8.57%
SVM (UMAP)	0.9889	0.5880	53.37%	< 0.000001	8.92%
Poisson (Dixon-Coles)	0.9845	0.5855	52.56%	< 0.000001	9.31%
Poisson (Dixon-Coles + Covariates)	0.9733	0.5780	53.74%	0.000410	10.47%
Poisson (Bivariate)	0.9678	0.5746	54.26%	0.030035	11.00%
Market (avg. bookmaker)	0.9607	0.5697	54.60%	—	11.76%
BET365	0.9615	0.5700	54.70%	—	11.71%

the correlation between home and away goals and incorporating relevant covariates leads to more accurate predictions. All the models tested fall short of the market odds, however, some more than others. The bivariate Poisson model’s *p*-value after the paired *t*-test is 0.030035, which is below the significance level of 0.05, indicating that its performance is statistically significantly different from BET365’s odds on the 95% confidence interval. However, conducting a *t*-test with a 99% confidence interval would not reject the null hypothesis that the log loss of the model and BET365’s odds come from a different distribution. This is a strong indication that the model is performing close to the market odds. Moreover, the BSS proves to be stably positive for all models meaning they comfortably beat the climatological probability baseline. The BSS of the bivariate Poisson model is 11.00%, which is only 0.76% lower than the market odds, thus providing another proof of the model’s strong performance.

8.1 Same configuration

8.2 Maximizing performance

9 Conclusion

None of the methods tested in this paper were on equal ground with the market odds for prediction of the final outcomes in EPL football matches despite the low disparity in the result metrics (0.0395 difference in log loss between the worst performing model (FFNN) and the market). The best performing model was the bivariate Poisson model with covariates, achieving just 0.0071 higher log loss than the market, thus being marginally worse in performance.

The results indicate that unsurprisingly, the market odds are highly accurate and efficient in predicting football match outcomes. Different machine learning models explored in this paper demonstrated varying levels of performance, with the bivariate Poisson model with covariates showing the most promise among the tested models. However, none of the models were able to surpass the market odds in terms of predictive accuracy.

10 Limitations and future research

As a multi-billion dollar industry, the sports betting market is highly competitive and efficient. These companies have an abundance of resources and data at their disposal, which allows them to create highly accurate odds.

The models tested in this paper were trained only on one sort of data, which is the historical data. This data is naive and in limited size. The models were trained on a dataset of 12 seasons of just a single league. Some machine learning models, such as neural networks or ensemble methods might suffer severely from the lack of

data, therefore also resulting in overfitting which had to be addressed with an unusually large regularization parameter. Naivety of the data is another substantial factor that could hinder the performance. Models trained in this paper were not exposed to any other matches throughout the season than the EPL itself. This means that they cannot recognize that a team might be fatigued because of a challenging match it has played just three days ago across the continent in a different competition.

While the historical data used for predictions indeed represents the core of any predictive model for such tasks, novel data including transfers of players, current line-ups, formations, injuries, coach changes, insider information or even weather conditions can all have a tiny impact on the performance by each on their own, however, adding this information on top of the historical data could possibly overcome the small but significant edge that the market has.

To sum up this section as well as the report itself, there are countless ways to conduct a research on this topic with plethora of different configurations, such as feature set(add-ons to the existing historical data), scale of the data (including more leagues), different machine learning methods, hyperparameter tuning, and evaluation metrics. The results of this paper are therefore not conclusive and should be treated as a starting point for further research in this area.

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